Parul University
Faculty of Engineering and Technology
Department of Applied Science & Humanities
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Subject: Quant and Reasoning (303105311)
Branch: CSE/ IT

TUTORIAL-8

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1. What is the speed of the train?
   I. The train crosses a signal pole in 18 seconds.
   II. The train crosses a platform of equal length in 36 seconds.
   III. Length of the train is 330 metres.
   A. I and II only
   B. II and III only
   C. I and III only
   D. III and either I or II only
   E. Any two of the three

2. What is the speed of the train?
   I. The train crosses a tree in 13 seconds.
   II. The train crosses a platform of length 250 metres in 27 seconds.
   III. The train crosses another train running in the same direction in 32 seconds.
   A. I and II only
   B. II and III only
   C. I and III only
   D. Any two of the three
   E. None of these

(6) Directions to Solve

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

1. At what time will the train reach city X from city Y?
   I. The train crosses another train of equal length of 200 metres and running in opposite directions in 15 seconds.
   II. The train leaves city Y and 7.15 a.m. for city X situated at a distance of 558 km.
   III. The 200 metres long train crosses a signal pole in 10 seconds.
   A. I only
   B. II only
   C. III only
   D. II and III only
   E. All I, II and III are required.

2. What is the length of a running train P crossing another running train Q?
   I. These two trains take 18 seconds to cross each other.
   II. These trains are running in opposite directions.
   III. The length of the train Q is 180 metres.

D.All I, II and III are required
E. Even with I, II and III, the answer cannot be obtained.

QUESTIONS RELATED TO AGES

(7) Directions to Solve

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question. Read the both statements and

- Give answer (A) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.
- Give answer (B) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.
- Give answer (C) if the data either in Statement I or in Statement II alone are sufficient to answer the question.
- Give answer (D) if the data even in both Statements I and II together are not sufficient to answer the question.
- Give answer(E) if the data in both Statements I and II together are necessary to answer the question.

1. What is Sonia's present age?
   I. Sonia's present age is five times Deepak's present age.
   II. Five years ago her age was twenty-five times Deepak's age at that time.
   A.I alone sufficient while II alone not sufficient to answer
   B.II alone sufficient while I alone not sufficient to answer
   C.Either I or II alone sufficient to answer
   D.Both I and II are not sufficient to answer
   E. Both I and II are necessary to answer

2. Average age of employees working in a department is 30 years. In the next year, ten workers will retire. What will be the average age in the next year?
   I. Retirement age is 60 years.
   II. There are 50 employees in the department.
   A.I alone sufficient while II alone not sufficient to answer
   B.II alone sufficient while I alone not sufficient to answer
   C.Either I or II alone sufficient to answer
   D.Both I and II are not sufficient to answer
   E. Both I and II are necessary to answer

3. Divya is twice as old as Shruti. What is the difference in their ages?
   I. Five years hence, the ratio of their ages would be 9 : 5.
   II. Ten years back, the ratio of their ages was 3 : 1.
   A.I alone sufficient while II alone not sufficient to answer

B. II alone sufficient while I alone not sufficient to answer
C. Either I or II alone sufficient to answer
D. Both I and II are not sufficient to answer
E. Both I and II are necessary to answer

(8) Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

(9) Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1. What is Arun's present age?
   I. Five years ago, Arun's age was double that of his son's age at that time.
   II. Present ages of Arun and his son are in the ratio of 11 : 6 respectively.
   III. Five years hence, the respective ratio of Arun's age and his son's age will become 12 : 7.
   A. Only I and II
   B. Only II and III
   C. Only I and III
   D. Any two of the three
   E. None of these

2. What is Ravi's present age?
   I. The present age of Ravi is half of that of his father.
   II. After 5 years, the ratio of Ravi's age to that of his father's age will be 6 : 11.
   III. Ravi is 5 years younger than his brother.
   A. I and II only
   B. II and III only
   C. I and III only
   D. All I, II and III
   E. Even with all the three statements answer cannot be determined.

3. What is the present age of Tanya?
   I. The ratio between the present ages of Tanya and her brother Rahul is 3 : 4 respectively.
   II. After 5 years the ratio between the ages of Tanya and Rahul will be 4 : 5.
   III. Rahul is 5 years older than Tanya.
   A. I and II only
   B. II and III only
   C. I and III only
   D. All I, II and III
   E. Any two of the three

(10) Directions to Solve

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

1. What will be the ratio between ages of Sam and Albert after 5 years?
   I. Sam's present age is more than Albert's present age by 4 years.
   II. Albert's present age is 20 years.
   III. The ratio of Albert's present age to Sam's present age is 5 : 6.
   A.Any two of I, II and III
   B.II only
   C.III only
   D.I or III only
   E. II or III only

2. What is the difference between the present ages of Ayush and Deepak?
   I. The ratio between Ayush's present age and his age after 8 years 4 : 5.
   II. The ratio between the present ages of Ayush and Deepak is 4 : 3.
   III. The ratio between Deepak's present age and his age four years ago is 6 : 5.
   A.Any two of I, II and III
   B. I or III only
   C. Any one of the three
   D. All I, II and III are required
   E. Even with all I, II and III, the answer cannot be obtained.

(11) Directions to Solve(Q. 1-3): Each of the equations below consists of a equations and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both statements and:

(a) If the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.
(b) If the data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
(c) If the data in statement I alone or in statement II alone are sufficient to answer the question.
(d) If the data in both the statement I and II are not sufficient to answer the question.
(e) If the data in both the statement I and II together are necessary to answer the question.

1. What is the average speed of a car?
I. The average speed of the car is 7 times the average speed of a truck where as the average speed of a bus is 49 km/hr.
II. The average speed of the truck is half of the average speed of a train where as the average speed of a Jeep is 60 km/hr.

2. What is the total cost of 5 kg.of Potato and 2 kg. of Mangoes together ?
I. The cost of 2 kg. of Potato is Rs 360 and 1 kg. of Mango is Rs 70.
II. The total cost of 5 kg. of Potato and 4 kg. of Mangoes is Rs 440.

3. What is the simple interest accrued on a principal of Rs 7600 in six years ?
I. The rate of the simple interest for the first 2 years is 5 pcpa.
II. The rate of the simple interest for the next 4 years is 8 pcpa

(12) Directions to Solve(Q. 1-5):

In each of these questions, one question is given followed by data in three statements I, II and III. You have to study the question and the data in statements and decide the question can be answered with data in which of the statements and mark your answeraccordingly.

1. What is the rate of interest pcpa?
Statements:
I. The difference between the compound interest and simple interest earned in two years on the amount invested is `100.
II. The amount becomes `19,500 in three years on simple interest.
III. The simple interest accrued in two years on the same amount at the same rate of interest is 3,000.
1) Only I and II
2) Only I and III
3) Only II and III
4) Only I and either II or III
5) None of these

2. What is the speed of the train in kmph?
Statements:
I. The train crosses an 'x' metre-long platform in 'n' seconds.
II. The length of the train is 'y' metres.
III. The train crosses a signal pole in 'm' seconds.
1) Any two of the three
2) Only II and III
3) Only I and III
4) All I, II and III
5) Question cannot be answered even with information in all three statements.

3. How many students passed in first class?
Statements:
I. 85% of the students who appeared in examination have passed either in first class or in second class or in pass class.
II. 750 students have passed in second class.
III. The number of students who passed in pass class is 28% of those passed in second class.
1) All I, II and III
2) Only I and III
3) Only II and III
4) Question cannot be answered even with information in all three statements.
5) None of these

4. What is the amount invested in Scheme 'B'?
Statements:

I. The amounts invested in Schemes ‘A’ and ‘B’ are in the ratio of 2 : 3.

II. The amount invested in Scheme ‘A’ is 40% of the total amount invested.

III. The amount invested in Scheme ‘A’ is `45,000.

1) Only I and II

2) Only I and III

3) Only II and III

4) All I, II and III

5) Only III and either I or II

5. What is the cost of flooring a rectangular hall?

Statements:

I. The length of the rectangle is 6 metres.

II. The breadth of the rectangle is two-thirds of its length.

III. The cost of flooring the area of 100 cm2 is `45.

1) Only I and III

2) Only II and III

3) All I, II and III

4) Question cannot be answered even with data in all three statements.

5) None of these